1. Hello, Everyone. This is Dr. Y. Zhang. In this session, we will learn cardiac impulse initiation, propagation and autonomic regulation. Here is my contact in case you need it.

2. Here are the session objectives. These are the contents that you are expected to learn.

3. First let's look at cardiac impulse generation or pacemaker activity. As you know, the normal heartbeat is initiated in the sinoatrial node (or sinus node) in the form of slow response action potentials. The action potentials generated in the sinus node will be transmitted to the rest of the heart.

4. First, let's look at the mechanism of pacemaker activity. The pacemaker activity or automaticity is generated by the phase 4 spontaneous depolarization. As you have learned already, slow response AP does not have a stable resting membrane potential. When slow response cell repolarizes, the member potential could drop to a level below -65mV. This is below the expected level of -60-65mV. This is called hyperpolarization. Hyperpolarization can trigger opening of the pacemaker channel. The pacemaker channel is also known as hyperpolarization-activated nucleotide-gated channel or HCN channel. Another name is the funny channel. Its current is the pacemaker current, funny current, or hyperpolarization-activated non-selective cation current. Note that the pacemaker channel can conduct both Na and K ions. The pacemaker channel is different from the fast Na+ channel seen in fast response fibers.

As illustrated here, you have to remember that pacemaker channel is the major source for phase 4 spontaneous depolarization. Besides the pacemaker channel, closing of the potassium channel IK1 and opening of the T-type Ca channel in later phase 4 also contribute to phase 4 spontaneous depolarization. So, the pacemaker activity is multifactorial.

5. In real life, the pacemaker activity is more complex. In addition to the mechanisms we have just discussed (pacemaker channel, T-type Ca channel, the potassium IK1 channel), marked by the green circle in the figure. It has been shown that periodic Ca release from the sarcoplasmic reticulum (SR) also contributes to pacemaker activity. Because the channels are located on the cell membrane, they are collectively called membrane clock or M-clock. The periodic Ca release mechanism is known as the Ca clock. You may find these terminologies in literature. So, it is good to know.

6. As a minimum you have to know that pacemaker current is the major force for pacemaker activity.

7. Now we will look at factors which could regulate the pacemaker activity. As we have learned, in order to generate an action potential, the membrane potential has to rise from its resting level to reach its threshold. Thus, there are 3 determinants here: the resting membrane potential level (or more correctly for the slow response action potential is the maximum diastolic potential level), the threshold level, as marked by the dashed-line in the figure, and the rate (or speed) of spontaneous depolarization.

You could think this way: The resting membrane potential level and the threshold level determine the distance. While the rate of spontaneous depolarization is the speed.

Thus, it is easy to understand that when the rate of spontaneous depolarization (or the speed) becomes slower, as shown here, the heart rate will become slower, assuming the other 2 factors are the same. Or verse versa.

8. This slide shows that if you can lower the resting membrane potential level, you basically increase the distance. Then it will need more time to travel from the resting potential to the threshold level, so the heart rate will be slower. The opposite is also true.

9. Here we will look at the threshold level. If you can increase the threshold level, this in effect increases the distance, thus, requiring more time from resting level to reach the threshold. In turn, the heart rate will be slower. On the other hand, if you can decrease the threshold level, resulting in a short distance, then the heart rate will be faster.

Now we have discussed the 3 major factors could affect the heart rates.

10. This slide shows the heart rates generated by the sinus node, the AV node and the His-Purkinje fibers. Obviously, the SA node generates the highest HR, normally in the range of 60-100 bpm in adults. That is why the sinus node is known as the primary pacemaker. The AVN can also generate Heart beats, but at a slower rate (about 40-60 bpm). The AVN is known as the secondary pacemaker (or backup pacemaker). When the sinus node fails, typically the AVN generates the HR.

You have to know that the His Purkinje fibers can also generate heat beats, although at a much slower rate, in the range of 20-40 bpm. This is because the His-Purkinje fibers also have small amount of pacemaker channels. Normally the His-Purkinje system do not have the chance to generate a heartbeat. They are overdriven by the heart beats coming from the sinus node. When both the SN and the AVN fail to generate heat beats, then you can see the pacemaker activity coming from the His Purkinje system, known as the ventricular escape beats.

11. Now we'll look at cardiac impulse propagation. This portion deals with how APs originated in the sinus node will be conducted to the rest of the heart. As illustrated here, AP generated in the sinus node will excite the atria, the AV node, the His bundle, bundle branches, Purkinje fibers and the ventricles in sequence. The detailed electrical propagation provides the basis for EKG formation, as you will learn in EKG lecture. It is also important in understanding the cardiac function.

12. Let's start from the sinus node (SN). The sinus node is located at the junction of superior vena cava (SVC) and the right atrium (RA). This region is commonly known as high RA clinically. At the center of the sinus node, there are nodal cells or pacemaker (P) cells. Between nodal cells and atrial myocytes there are transitional cells. Action potentials generated in the sinus node, going through the transitional cells, activates nearby atrial myocytes. Inside the sinus node, there are low-conductance connexins (CX 45), so the conduction inside the sinus node is slow.

Normal sinus rate (HR) in adults is in the range from 60-100bpm, on average 72 bpm. If the HR is above 100bpm, it is called sinus tachycardia. If it is <60 bpm, it is called sinus bradycardia.

13. Now we will look at electrical propagation inside the atria: Since the sinus node is located at the high-right atrium, the high RA region is the first to be activated. Then the atrial excitation goes downward to the AV node and leftward to the left atrium.

Within RA, the activation goes from the sinus node down to the atrioventricular node, preferentially going through 3 internodal pathways or tracts, as marked in the figure. There are anterior, lateral and posterior inter-nodal pathways. These pathways are consisted of normal atrial myocytes, not the specialized fibers.

The activation from right atrium to left atrium is preferentially going through a muscle tract, called Bachman's bundle, located on the roof (top) of both atria. Thus, you should know that the RA is activated earlier than the LA.

From here you should know that generally the atrial depolarization direction is from top to the bottom and from right to the left. This excitation sequence determines the P wave axis you will learn in EKG lecture.

14. This slide shows that activation along the muscle bundles (or longitudinal conduction along the fiber orientation) is faster than across the fiber orientation (or transverse conduction). Note that there are more gap junctions between longitudinally aligned fibers (end to end connection) than across fibers (or side connection), and there are fewer cells to travel longitudinally for a given distance.

15. This slide just shows the gap junction between cells. Gap junctions are clusters of intercellular channels that allow direct diffusion of ions and small molecules between adjacent cells. Gap junctions are made of different connexins (Cxs 40, 43, 45 etc) with different conductance.

16. Atrial depolarization creates the P wave on EKG, as shown in the right-side figure, top trace. The middle trace is the intracardiac electrogram recorded from high right atrium, near the sinus node. Timing-wise the atrial signal recorded from HRA happens at the beginning of the P wave on EKG. The bottom trace is the intracardiac electrogram recorded near the His bundle. It contents the distal atrial electrogram (A), the His bundle electrogram (H) and ventricular electrogram (V). These intracardiac electrograms are recorded using special catheters inserted into the heart.

Remember the atrial activation creates a downward and leftward average vector. The vector determines the “P wave” axis on EKG. This is why in some leads the P wave should be positive and in some leads the P wave should be negative. And you will learn this in EKG lecture.

17. From atrial excitation, you should know that atrial contraction starts from top going down, moving the blood from the atria down into the ventricles.

18. When the excitation from atria reaches the AVN, it will take time to pass the AVN, because the conduction inside the AVN is very slow.

The AV node is the only normal electrical connection between the atria and the ventricles.

The AV node is located in the lower interatrial septum. Histologically, the AV node can be divided into central region, the nodal cells (N), and two transitional regions (between atrial and node, the AN cells and the other from node to the His bundle, the NH cells.

Clinically, the AVN is known in an area termed as the triangle of Koch. The triangle is demarcated by the tricuspid annulus, tendon of Todaro, and the opening of the coronary sinus, as indicated in the figure. The AV node is located at the apex of this triangle.

The AVN plays 3 major roles. Normally, the AV node creates a physiological AV delay in transmitting activation from the atria to the ventricles. This delay (about 0.1s) is importance because it allows the atria to complete their contraction, filling the ventricles before ventricular contraction starts. Thus, the physiological AV delay is needed to optimize the ventricular filling before its contraction. It is estimated that about 20% of ventricular filling is contributed by the atrial contraction.

The 2nd role is that the AV node serves as a secondary or backup pacemaker in case the sinus node fails. Its rate is slower, about 40-60 bpm.

The 3rd role is that the AV node can serves as a filter during atrial tachy-arrhythmias. For example, during atrial fibrillation, the atrial rates could be 400-600bpm. If all atrial beats were conducted, ventricular fibrillation would occur. Thanks to the AV node, many of the atrial impulses are blocked in the AVN, resulting in a relatively slower ventricular rates (typically in the range of 100-150bpm) during AF.

AVN activation is not shown on EKG.

19. This slide shows again the triangle of Koch, in an anatomically correct orientation.

20: Here I am listing the major reasons why conduction is very slow in the AVN.

First, the AVN generates slow response AP, with lower dv/dt (maximal speed of depolarization) & lower amplitude. The AV node contains low-conductance connexin (Cx45). The AV node fibers are small in size. The AV node has dense connective tissue

All these contribute to the slow AVN conduction.

Here is a sample question. In order to inhibit AV conduction, which medication

would you like to choose? L-type calcium channel blocker or sodium channel blocker?

And you should know the answer.

21. After the AVN, excitation reaches the His-Purkinje system. The His bundle is located in the central fibrous body and upper interventricular septum. The His bundle bifurcates to form the right and the left bundle branches. The right bundle branch travels down along the septum and supplies Purkinje fibers to the right ventricle. The left bundle branch shortly bifurcates again to form two major branches called the left anterior fascicle and the left posterior fascicle. For the EKG purpose, you have to know that the left bundle branch also gives off a small branch called septal fascicle. Due to this small fascicle, the upper left interventricular septum is the first to be activated in the ventricles. We will discuss this later.

The His bundle, the right and left bundle branches are insulated in the fibrous tissue. They do not have direct contact with ventricular myocytes. When both left and right bundle branches reach the apex, they start giving off the Purkinje fibers sub-endocardially. The Purkinje fibers have direct contact with ventricular myocytes and can excite the ventricles. This is why the endocardial ventricular myocytes are the first to be activated.

Remember, the Purkinje fibers have the highest conduction velocity in the heart and are responsible for synchronized ventricular activation and contraction.

22. The activation of the His-Purkinje system is not shown on EKG. However, His electrogram can be recorded using intracardiac catheter, as shown in the Figure marked with red circle. Left or right bundle branch block can be diagnosed with EKG, because BBB alters ventricular activation sequence. You will learn this in EKG lecture.

23. Here are the reasons why the Purkinje fibers have the highest CV.

First, Purkinje fibers generate fast response AP, with high dv/dt (maximal speed of depolarization) & high amplitude. Second, they have largest cell size. Third, they have high-conductance connexin (Cx 43). That’s why the Purkinje fibers have the highest CV.

24. Now let’s look at ventricular activation. Please look at this figure and try to find the earliest activation site in the ventricles. It is here, at the upper interventricular septum on the left side, marked with a star. As mentioned already, this activation is due to the septal branch from the left bundle branch. This activation creates a rightward vector. This vector is responsible for the beginning portion of the QRS complex, so-called “septal q wave” on the left-side EKG leads. You will learn this in EKG lecture.

If you look at the numbers, the general activation of the ventricles is from apex to base, and from endocardium to epicardium.

25. Here I am giving you another sample question: Which side of the ventricles is activated earlier: endocardial side vs epicardial side? and why? Now You should be able to answer this question.

26. As already discussed, the earliest ventricular activation site is at the upper inter-ventricular septum on the left side, which creates a rightward vector (marked with arrow), the remaining general activation of the ventricles is from apex toward base and from endocardial fibers to epicardial fibers, due to the Purkinje fibers' location and distribution.

The ventricular depolarization produces the QRS complex on EKG (marked with a circle in the figure).

27. Here is another sample question: Which medication can slow ventricular conduction? L-type calcium channel blocker or sodium channel blocker? You should know this now.

28. Here is the functional importance of ventricular excitation: Ventricular contraction starts from apex toward base, pumping blood up into the aorta. Also notice that contraction starts from the endocardial layer (the red line) and spread to the epicardial layer (the blue line). Think this way, if the contraction starts from outside while the inside wall is still relaxed, it would be counter-productive.

29. After ventricular depolarization, here comes ventricular repolarization. You have seen this figure before. Here I am not going to ask you how K concentration affects AP. I would like to ask you to look carefully at the action potential duration. Clearly the AP duration of epicardial myocytes is shorter than mid-layer or endocardial myocytes. The shorter AP duration means the cell repolarizes earlier (or faster). From here, you should know ventricular repolarization starts from epicardial side. This is due to the differences in the density and kinetics of the potassium channels involved in repolarization. For example, the epicardial myocytes have higher density of IK_{to} (the transient outward K channels).

30. Here are the ventricular repolarization sequence: Ventricular repolarization is from epicardial to endocardial (this is opposite to ventricular depolarization), and from apex to base (this is the same direction as depolarization). Due to these changes, ventricular depolarization and repolarization vectors are generally in the same direction.

On EKG, ventricular repolarization creates the T wave. Because ventricular repolarization vector points to the similar direction as the ventricular depolarization, normally the T wave polarity should follow the QRS complex polarity. In other words, if the QRS is positive, the T wave should also be positive, or verse versa. You'll learn this in EKG lecture.

As shown on the EKG trace, the atrial depolarization, the ventricular depolarization and repolarization create the 3 major EKG waveforms (the P wave, the QRS complex and the T wave).

31. The ventricular repolarization sequence also has functional importance: Repolarization starts at apex, epicardially (outside wall) to optimize ventricular relaxation. You can think this way, ventricles cannot dilate unless the outside wall is relaxed.

32. Here is the concept of conduction velocity: CV is the speed at which APs are propagated in cardiac tissue. CV differs in different parts of the heart, as shown in the table. The Purkinje fibers have the fastest CV. The CV is the slowest in the AVN, with atria and ventricles in between.

These are the factors could affect CV, including AP types, cell size, gap junction type (high vs low-conductance), and fibrosis contents.

33. Here are the minimum requirements you have to know.

Note that First aid listed the atrial CV is faster than the ventricular. This is probably due to different sources.

34. Now let's look at the autonomic effects on the heart.

As you know, the autonomic nervous system has 2 branches: sympathetic nerve and parasympathetic nerve or vagal nerve. As summarized here, sympathetic nerve stimulation increases heart rate, increases AV conduction and increases myocardial contractility.

And generally vagal nerve has opposite effects.

35. Vagal stimulation can decrease heart rate. You should know that vagal nerve releases its neurotransmitter acetylcholine (Ach). Ach binds to muscarinic (M) - receptors. This leads to activation of acetylcholine sensitive K+ channels. This is a ligand gated channel, different from previously discussed voltage-gated K channels. Activation of acetylcholine sensitive K channel generates an outward K+ current (Ik_{ach}), resulting in a more negative membrane potential (or hyperpolarization). Acetylcholine can also reduce pacemaker current and calcium current, thus increasing the time needed for spontaneous depolarization, and slowing the heart rate. Slowing of the heart rate is termed negative chronotropic effect.

The sinus node and the AV node are densely innervated by the vagal fibers. Strong VS can inhibit both the sinus node and the AV node. In this case, other potential pacemaker sites in the ventricles, like the Purkinje fibers, could generate action potential, but the heart rate is very slow, <40bpm. This is called ventricular escape beat.

36. Here is the signaling pathway mediating the vagal effects. Vagal neurotransmitter acetylcholine binds to its M-receptor, activating the inhibitory G-protein. This leads to the dissociation of βγ-subunits from the α-subunit. In this case, the βγ-subunits of the G-protein are the activating agent. They bind to and open the acetylcholine sensitive K+ channels, hyperpolarizing the cell. The inhibitory α-subunit also decreases adenylyl cyclase, reducing cAMP level and inhibiting pacemaker channel and L-type Ca channel, resulting in HR slowing.

Vagal stimulation decreases HR, it's termed negative chronotropic effect.

Decrease AV conduction, termed negative dromotropic effect.

And Decrease myocardial contractility, termed negative inotropic effect.

37. Now Let's look at the sympathetic nerve. Activation of sympathetic nerve, like during exercise, or in "fight or flight" conditions, increases the heart rate. Excitation of sympathetic nerve releases its neurotransmitter norepinephrine. NE binds to adrenergic β1 receptors in the heart. This eventually leads to increased pacemaker current and Ca current, thus increase HR and contractility.

There is a concept called Maximum Heart Rate. The maximum HR is the heart rate could be achieved under maximal sympathetic stimulation (for example, during exercise testing). Maximum HR is age-dependent. It decreases as people age. The rate is estimated as 220bpm-age (in years). For example, for a 20-year old, the maximum HR is about 200bpm.

38. Here is the signaling pathway mediating the sympathetic effects. NE binds and activates the β1 receptor in the heart, and subsequently activates the stimulating Gs-protein. In this case, the α-GTP is the active agent. It binds to and activates adenylyl cyclase, which converts ATP into cAMP, leading to activation of pacemaker channel and activates protein kinase A and subsequently the L-type calcium channel and calcium release channel, or the ryanodine receptor. Thus, it increases HR, AV conduction, and cardiac contractility, known as the positive chronotropic effect, dromotropic effect, and inotropic effect, respectively.

39. Here is the concept of intrinsic heart rate. Intrinsic heart rate is the heart rate when you block both sympathetic and vagal effects. In other words, without autonomic influence. The figure shows the IHR in male (up) and female (bottom). IHR is age-dependent. As people age, IHR decreases. In general, you can see, the IHR is around 100bpm in adults. Considering the resting HR in adults is about 70bpm, here is the question, in resting state, which one is dominant, the sympathetic nerve or parasympathetic nerve?

The correct answer is the vagal effect is dominant in resting state. It brings down the HR from about 100bpm to 70bpm.

40. Here is the summary of this session. These are the contents you are expected to master.

41. Finally, here is the link for your feedback. We appreciate any feedback you might have. Thank you for your time and attention.